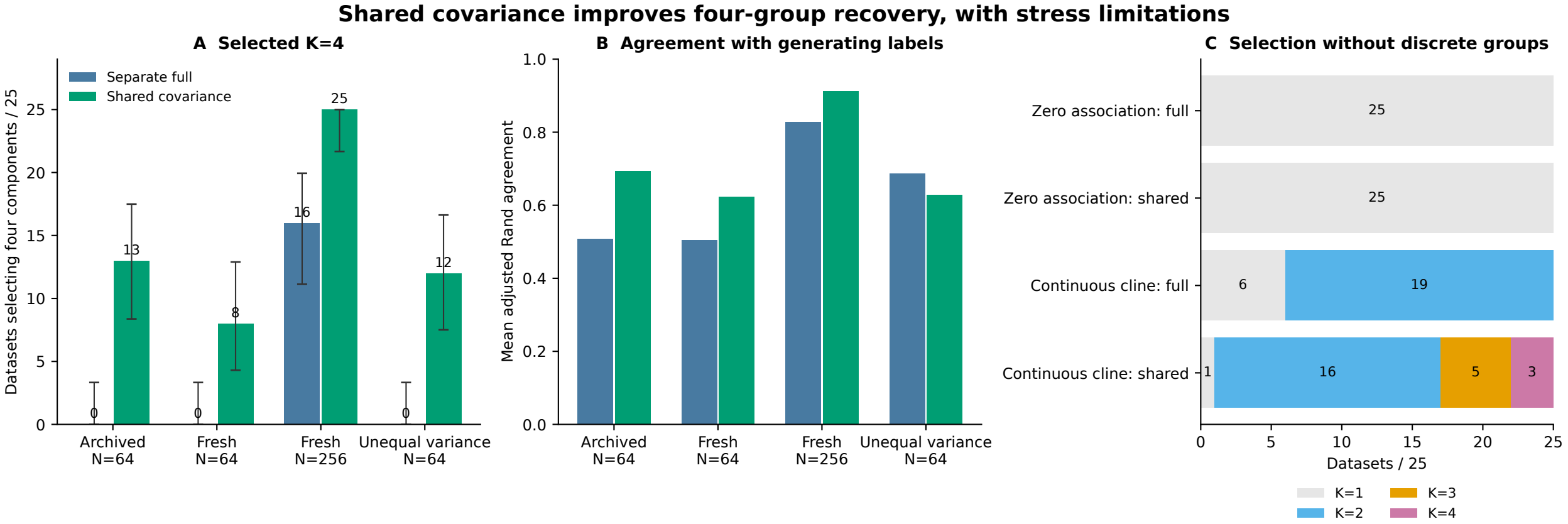
Shared covariance improves four-group recovery, with stress limitations
A Selected K=4
Separate full
Shared covariance
Datasets selecting four components / 25
0
5
10
15
20
25
0
13
0
8
16
25
0
12
Archived N=64
Fresh N=64
Fresh N=256
Unequal variance N=64
B Agreement with generating labels
Mean adjusted Rand agreement
0.0
0.2
0.4
0.6
0.8
1.0
Archived N=64
Fresh N=64
Fresh N=256
Unequal variance N=64
C Selection without discrete groups
Zero association: full
25
Zero association: shared
25
Continuous cline: full
6
19
Continuous cline: shared
1
16
5
3
0
5
10
15
20
25
Datasets / 25
K=1
K=2
K=3
K=4
Frozen two-PC settings; pointwise Wilson 95% intervals in A. Shared covariance remains opt-in.